

EVIDENCE ROOM

AP Agent Stack Overview

Sixteen agents. One operating system.

Tier 2 · Professional

Operating Evidence · Not legal, tax, or accounting advice

Agent Stack Overview — AP Agent OS Professional

Product: Evidence Room — AP Agent OS Professional **Artifact:** Agent Library · Architecture Overview
Audience: CFO, Controller, Head of AP / Shared Services, Finance Transformation **Version:** 1.0

1. Purpose of this document

This overview defines how the sixteen AP agents form a coherent operating system across invoice-to-pay — what each agent owns, how work moves between them, where humans remain accountable, and how the stack stays ERP-agnostic.

Use it as the architectural reference before configuring individual agent charters (A01–A16), the responsibility model (L0–L4), and the exception taxonomy.

2. Design principles

Human accountability first	Agents earn scope. Full autonomy is never the default. Every material AP decision has a named human owner.
Payment authorisation stays human	No agent may release, approve, or execute payment. Payment Proposal Review prepares and flags; treasury / AP leadership authorises.
Control before speed	Throughput gains that weaken segregation of duties, audit trail, or exception clarity are rejected.
ERP-agnostic by construction	Agents consume and produce normalised AP objects (invoice, PO, receipt, vendor, coding, approval, payment proposal). ERP-specific connectors are implementation detail, not product identity.
Exception-first operations	Clean straight-through work is measured; value concentrates on classified, owned, aged exceptions.
Evidence over narrative	Every material recommendation or action leaves durable evidence: inputs used, rule applied, confidence, human decision, timestamp.

3. The sixteen-agent stack

Agents are numbered for governance and KPI mapping, not for rigid sequential batching. Many run in parallel; the Orchestrator sequences handoffs and escalations.

A01	Invoice Intake	Capture, classify, and register inbound invoices into a controlled work queue

A02	Invoice Validation	Verify completeness, coding readiness, tax fields, and supplier identity signals
A03	Matching	Perform PO / non-PO match logic against PO, receipt, and invoice lines
A04	Exception Triage	Classify exceptions, assign owners, set SLA clocks, route work
A05	Goods Receipt	Chase and reconcile receipt evidence for receipt-required invoices
A06	PO Quality	Detect and prevent PO defects that create downstream match failures
A07	Approval	Drive approval routing within DOA; never invent authority
A08	Supplier Resolution	Coordinate supplier queries with configurable human approval gates
A09	Internal Follow-Up	Chase requesters, buyers, and GR clerks with evidence packs
A10	Duplicate & Anomaly	Surface duplicate and anomaly candidates — not a fraud guarantee
A11	Vendor Statement Reconciliation	Reconcile statements to open items and propose discrepancy packs
A12	Payment Proposal Review	Prepare payment batches for human authorisation only
A13	AP Close	Support period-end completeness, accruals signals, and aged WIP
A14	AP Reporting	Produce operating dashboards and exception analytics
A15	Root Cause	Aggregate exception patterns into preventable causes and actions
A16	AP Manager / Orchestrator	Prioritise, sequence, escalate, and govern the agent fleet

4. How agents interact

4.1 Core invoice flow (happy path)

` Inbound channel → A01 Invoice Intake → A02 Invoice Validation → A03 Matching → [clean match] → A07 Approval (if required) → A12 Payment Proposal Review (batch prep) → Human payment authorisation (outside agent autonomy) `

4.2 Exception path

` A02 / A03 / A10 / A11 detect defect → A04 Exception Triage (classify, own, age) → specialised resolver: A05 Goods Receipt A06 PO Quality A07 Approval A08 Supplier Resolution A09 Internal Follow-Up A10 Duplicate & Anomaly (investigation pack) → resolution evidence returned to A04 → invoice returns to A02/A03 or proceeds to A07/A12 `

4.3 Analytical / period loop

` A04 exception ledger + ERP open items → A11 Vendor Statement Reconciliation → A13 AP Close → A14 AP Reporting → A15 Root Cause → A16 Orchestrator (capacity, pilot scope, promotion/demotion signals) `

4.4 Orchestrator role (A16)

The AP Manager / Orchestrator does **not** replace the Head of AP. It:

- Maintains the workboard: priority queues by value, age, SLA breach risk, and control severity.
- Routes tasks to specialised agents within configured autonomy levels.
- Enforces stop-conditions (auth failure, control break, ambiguous payment outcome, policy gap).
- Surfaces promotion/demotion evidence for the responsibility model.
- Escalates to humans when confidence, amount, vendor risk, or policy requires it.

Humans set policy, approve scope changes, authorise payments, and own residual risk.

5. Human-in-the-loop principle

Every agent operates inside an explicit autonomy band (L0–L4). Promotion is earned with evidence; demotion is automatic on control failure.

Always human-owned:

- Payment authorisation and bank file release
- Master-data banking changes (final approve)
- DOA exceptions and override of approval policy
- Write-offs, credit settlements above threshold, disputed legal positions
- Agent autonomy promotion beyond L2 (typical gate; organisations may set stricter)
- Any action that would post irreversible financial impact outside pre-agreed guardrails

Typically agent-assisted first:

- Intake classification and duplicate *candidates*
- Match suggestions and variance explanations
- Exception classification and chase drafts
- Statement discrepancy packs
- Close checklists and reporting drafts

The operating rule: *agents prepare; humans decide on material risk; earned autonomy expands only where error rates and controls permit.*

6. ERP-agnostic integration pattern

6.1 Canonical objects

Integrate via a thin **AP object layer**, not ERP screens:

Invoice header/lines	Supplier, entity, dates, amounts, tax, currency, PO refs, attachments

Purchase order	Status, lines, prices, qty, remaining, buyer, entity
Goods receipt	Receipt ID, qty, date, PO line link, receiver
Vendor master	Legal name, VAT/tax IDs, remit-to, bank tokens (masked), hold flags
Coding / dimensions	Cost centre, GL, project, tax code
Approval / DOA	Approver, amount band, status, timestamp
Payment proposal	Invoice IDs, net amount, pay date, method, hold reasons
Exception record	Code, owner, age, status, evidence links

6.2 Connector posture

- **Read-heavy first.** Prefer read APIs / exports for observation and recommendation (L0–L1).
- **Write with guardrails.** ERP writes (status, coding suggestion accept, park codes) only at L2+ with idempotency keys and dry-run mode.
- **No dual auth families** where the platform forbids them; follow each ERP's auth model exactly.
- **Environment awareness.** Demo/sandbox vs production credentials never mixed.
- **Idempotent outbound actions.** Re-sends after timeout must not create duplicate ERP documents.

6.3 Compatible estates

Designed to sit across Dynamics 365, SAP (ECC/S/4), Oracle Cloud/EBS, NetSuite, Workday Financials, and mid-market ERPs with AP modules — provided invoice, PO, receipt, vendor, and payment proposal data can be read (and optionally written) through API, RPA with controls, or controlled file exchange.

Evidence Room does **not** replace the ERP, OCR vendor, or payment bank gateway. It governs how agents use them.

7. Control and evidence spine

Across the stack, every material agent output must carry:

- 1 **Trace ID** linking invoice / exception / batch
- 2 **Inputs referenced** (document versions, master-data snapshot IDs)
- 3 **Rule or model version**
- 4 **Confidence / match quality** where applicable
- 5 **Proposed action vs executed action**
- 6 **Human decision** (approve / reject / amend) with actor and time
- 7 **Retention** aligned to audit policy

A14 Reporting and A16 Orchestrator consume this spine for KPI and promotion gates.

8. Suggested sequencing for first deployment

Do not deploy all sixteen at once.

Pilot wave (typical): A01 Intake → A02 Validation → A03 Matching → A04 Exception Triage → A10 Duplicate & Anomaly (candidate-only) → A16 Orchestrator (light).

Second wave: A05, A09, A07, A14.

Third wave: A06, A08, A11, A12 (prep-only), A13, A15.

Expand autonomy only after shadow and controlled pilot gates in `RESPONSIBILITY_MODEL.md`.

9. Related artifacts

Responsibility model L0–L4	`Agent_Library/RESPONSIBILITY_MODEL.md`
Agent charters A01–A16	`Agent_Library/A01_...` through `A16_...`
Exception taxonomy	`Controls/01_EXCEPTION_TAXONOMY.md`

Evidence Room — AP Agent OS Professional. Operating layer for AP agents. Not an ERP, not a payment system, not a fraud guarantee.